

Phishing Simulation & Detection

I built a safe GoPhish campaign in my own lab, sent it to two test mailbox I own, then turned around and detected the activity in Splunk.

Build - Launch - Compromise - Detect

Swipe to see how →

I split this lab into two halves.

First, I built the campaign in GoPhish on it's own

NAT'd from my Ubuntu VM

Then I used Splunk to detect what happened.

That made the logs easier to understand because

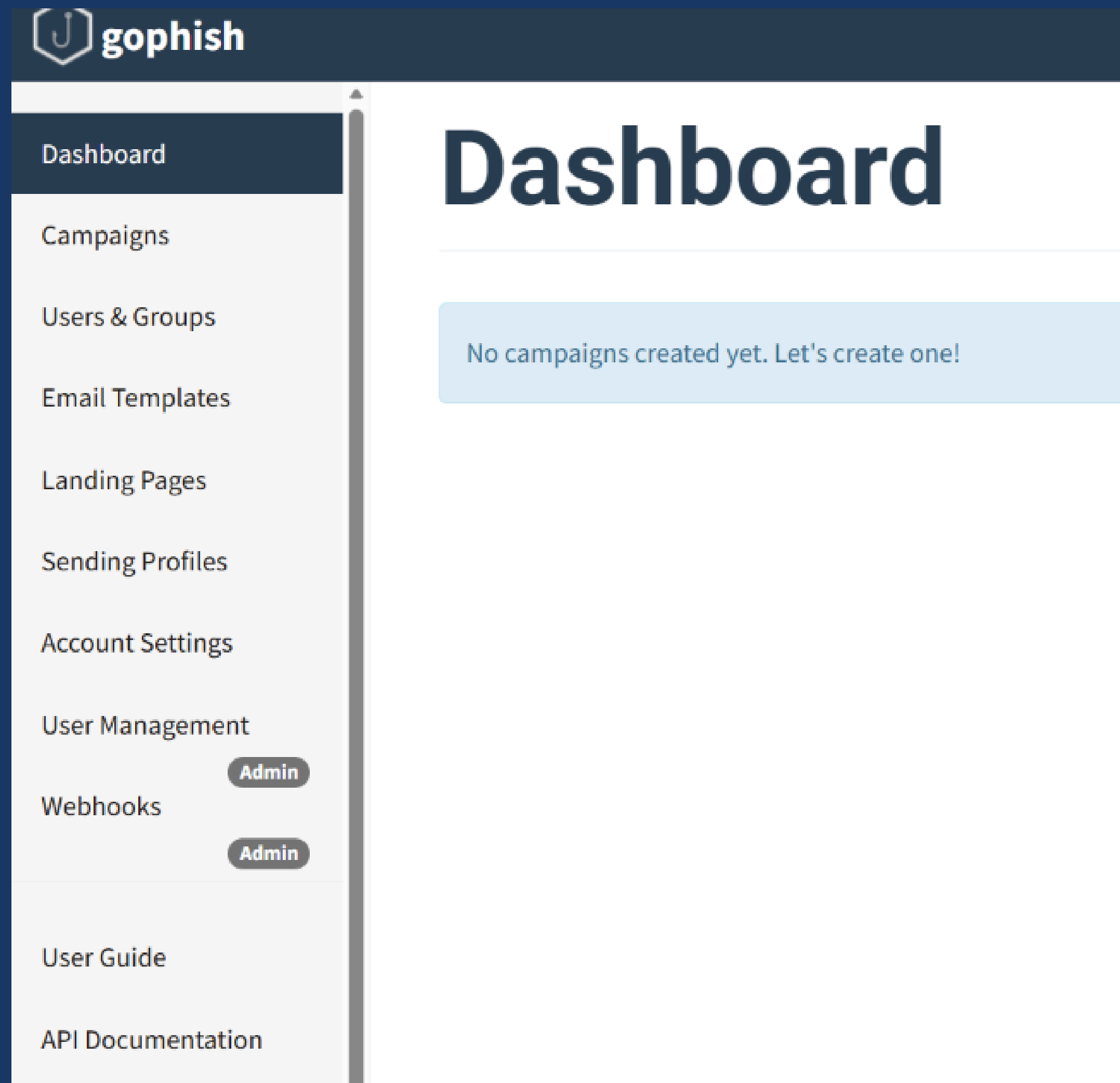
I already knew the truth.

Then I wanted to prove it with evidence.

From one dashboard, I built the sender, the lure, the fake landing page, and the target list.

Campaign: **“Password Reset Simulation.”**

A password-reset lure is boring on purpose. Boring is what people click without thinking.



Before building the full campaign, I tested whether GoPhish could actually send mail through Gmail SMTP. The test email worked, which meant the sending profile was good.

Default Email from Gophish



seanwhite.soclab@gmail.com

To: me · Fri, Jul 3 at 3:45 PM ▾

It works!

This is an email letting you know that your gophish configuration was successful.

Here are the details:

Who you sent from: seanwhite.soclab@gmail.com

Who you sent to:

Now go send some phish!

I analyzed tracking pixels and suspicious links as a defender.

Here, I built them in a safe lab.

The tracked link logs the click.

The 1x1 pixel logs the open.

I also ran into a GoPhish tracking-image issue, so I added the tracking pixel manually and validated it later.

```
<p>Hello,</p>

<p>Our records indicate that your account password is set to expire within 24 hours. To avoid disruption to your account access, please reset your password using the secure link below.</p>

<p><a href="{{.URL}}">Reset My Password</a></p>

<p>If you do not reset your password before it expires, you may be temporarily locked out of your account until IT support can assist you.</p>

<p>Thank you,<br>
IT Support Team</p>
```

This was a fake password reset page I built.

Not a clone of a real company.

Not used on real people.

Only fake test credentials.

The important part was seeing where “submitted” gets logged.

GoPhish can capture whatever gets typed into the form, which is exactly why the lab rule matters:

```
<html>
<head><title>Account Password Reset</title></head>
<body style="font-family: Arial, sans-serif; text-align: center; margin-top: 80px;">
  <h2>Reset Your Password</h2>
  <p>Please enter your credentials below to continue.</p>
  <form method="post">
    <input type="text" name="username" placeholder="Username" style="padding:8px; width:250px; margin-bottom:10px;"><br>
    <input type="password" name="password" placeholder="Password" style="padding:8px; width:250px; margin-bottom:10px;"><br>
    <button type="submit" style="padding:8px 20px;">Reset Password</button>
  </form>
</body>
</html>
```

Credentials submitted: **17:19:32**

One minute from inbox to captured login.
No malware. No exploit. Just a believable
email and one form.

That is why phishing still works. It does
not beat your defenses. It skips them.

Sent.

Opened.

Clicked.

Submitted.

One test target.

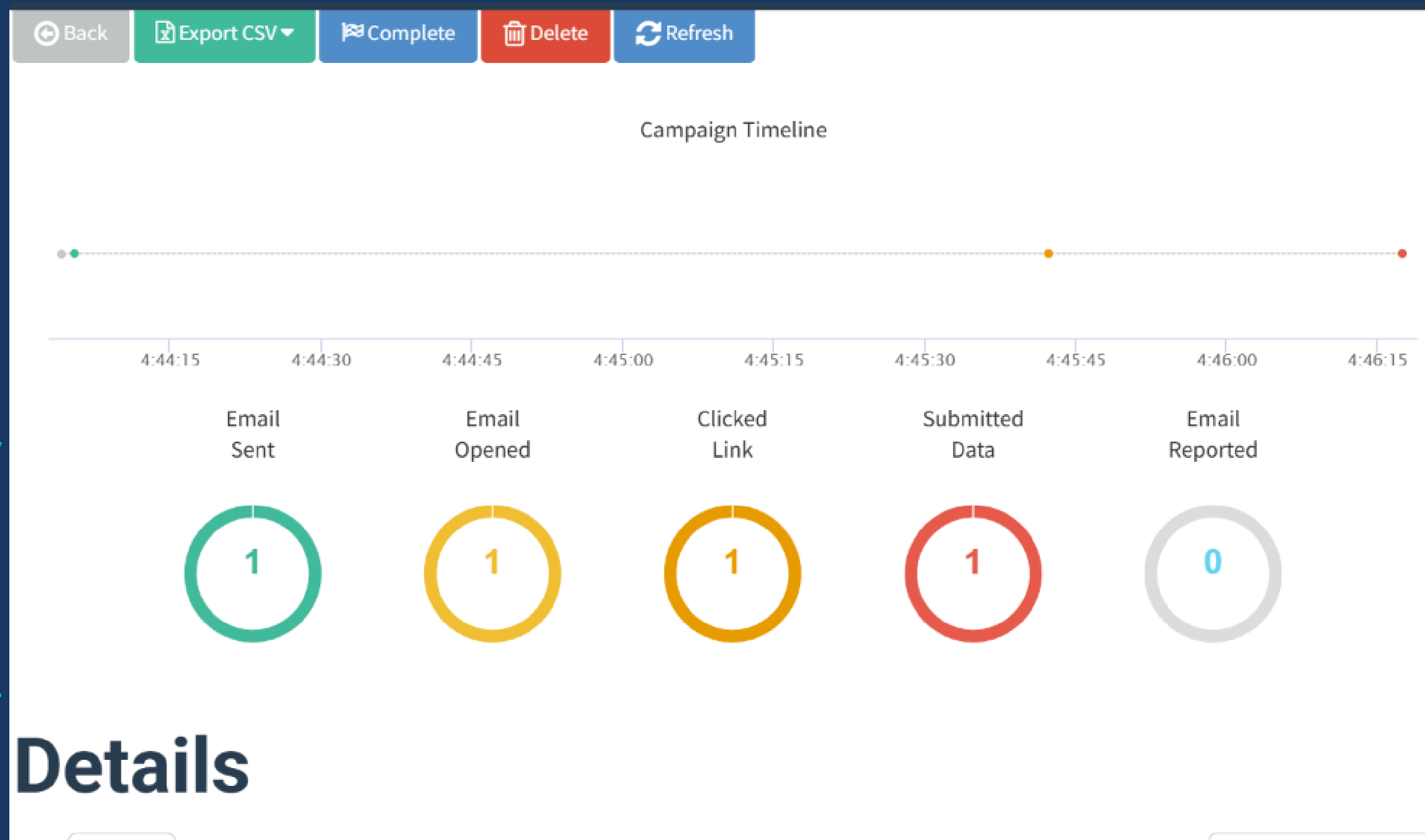
One controlled
campaign.

One complete chain.

Watching the activity

move through

GoPhish made the
detection side much
easier to understand.



The first send went to spam.
Honestly, that was a good finding.

A brand-new sender with no reputation should get filtered.
I kept this in the lab because it matters.

Spam filtering partially worked.
That is not something to hide.
That is a control doing part of its job.

Default Email from Gophish 📁 Inbox



seanwhite.soclab@gmail.com

To: me · Fri, Jul 3 at 3:45 PM ▾

It works!

This is an email letting you know that your gophish configuration was successful.

Here are the details:

Who you sent from: seanwhite.soclab@gmail.com

Who you sent to:

Now go send some phish!

The raw GoPhish logs had noise from my own admin activity.

The useful events were much smaller:

Email sent.

Opened/clicked.

Submitted.

Format ▾ Show: 20 Per Page ▾ View: List ▾		
i	Time	Event
>	7/3/26 5:19:32.000 PM	time="2026-07-03T17:19:32-07:00" level=info msg="192.168.220.1 - - [03/Jul/2026:17:19:32 -0700] \"POST /?rid=JZgWFhx HTTP/1.1\" 302 0 \"http://192.168.220.128/?rid=JZgWFhx\" \"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36 Edg/150.0.0.0\"" host = ubuntu-lab source = /home/sean/gophish/gophish_access.log sourcetype = gophish_access
>	7/3/26 5:19:25.000 PM	time="2026-07-03T17:19:25-07:00" level=info msg="192.168.220.1 - - [03/Jul/2026:17:19:25 -0700] \"GET /?rid=JZgWFhx HTTP/1.1\" 200 615 \"\" \"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36 Edg/150.0.0.0\"" host = ubuntu-lab source = /home/sean/gophish/gophish_access.log sourcetype = gophish_access
>	7/3/26 5:18:31.000 PM	time="2026-07-03T17:18:31-07:00" level=info msg="192.168.220.1 - - [03/Jul/2026:17:18:31 -0700] \"GET /api/campaigns/3/results?{} HTTP/2.0\" 200 654 \"https://192.168.220.128:3333/campaigns/3\" \"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36 Edg/150.0.0.0\"" time="2026-07-03T17:18:32-07:00" level=info msg="Email sent" email="\"Cali Test\" <calipartyboy56@yahoo.com>" envelope_from="\"IT Support\" <seanwhite.soclab@gmail.com>" smtp_from=seanwhite.soclab@gmail.com host = ubuntu-lab source = /home/sean/gophish/gophish_access.log sourcetype = gophish_access

This was the number I cared about most.
How long did it take to go from email sent to credentials submitted?
Splunk tied the activity together and calculated the timeline.
Around 60 seconds.
That is why phishing is dangerous.
The whole chain can happen fast.

New Search

Save As ▼

Create Table View

Close

```
sourcetype=gophish_access "Email sent" OR "rid="
| rex field=_raw "rid=(?<rid>\w+)"
| rex field=_raw "\"(?<method>GET|POST) "
| rex field=_raw "email=\"[^\"]*<(?(sent_email>[^\"]+)>\""
| eval stage=case(match(_raw, "Email sent"), "Sent", method=="GET", "Opened/Clicked", method=="POST", "Submitted")| eval target_email
=coalesce(sent_email, "calipartyboy56@yahoo.com")| stats earliest(_time) as first_seen, latest(_time) as last_seen, values(stage) as
stages by target_email
| eval time_to_compromise=last_seen-first_seen
| convert ctime(first_seen) ctime(last_seen)
| table target_email, stages, first_seen, last_seen, time_to_compromise
```

Time range: All time ▼



✓ 3 events (before 7/3/26 7:16:11.000 PM)

No Event Sampling ▼

Job ▼



Smart Mode ▼

Events

Patterns

Statistics (1)

Visualization

Show: 20 Per Page ▼

Format ▼



Preview: On

target_email ↕	stages ↕	first_seen ↕	last_seen ↕	time_to_compromise ↕
calipartyboy56@yahoo.com	Opened/Clicked Sent Submitted	07/03/2026 17:18:31	07/03/2026 17:19:32	61

I did not just trust the dashboard.
I checked the aggregate against the raw logs.
Splunk grouped the activity, but I still wanted to confirm the timing and recipient ID myself.
That was one of the best lessons from the lab:
Never quote a number just because the tool gave it to you.
Validate it.

New Search

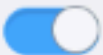
Save As ▾ Create Table View Close

sourcetype=gophish_access "Email sent" OR "rid=" | rex field=_raw "rid=(?<rid>\w+)" | table _time, rid, _raw | sort _time

Time range: All time ▾  

✓ 3 events (before 7/3/26 7:10:03.000 PM) No Event Sampling ▾ Job ▾       Smart Mode ▾

Events Patterns **Statistics (3)** Visualization

Show: 20 Per Page ▾  Format ▾  Preview: On

_time ▾	rid ▾	_raw ▾
2026-07-03 17:18:31		time="2026-07-03T17:18:31-07:00" level=info msg="192.168.220.1 - - [03/Jul/2026:17:18:31 -0700] \"GET /api/campaigns/3/results?{} HTTP/2.0\" 200 654 \"https://192.168.220.128:3333/campaigns/3\" \"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36 Edg/150.0.0.0\""
		time="2026-07-03T17:18:32-07:00" level=info msg="Email sent" email="\"Cali Test\" <calipartyboy56@yahoo.com>" envelope_from="\"IT Support\" <seanwhite.soclab@gmail.com>" smtp_from=seanwhite.soclab@gmail.com
2026-07-03 17:19:25	JZgWFhx	time="2026-07-03T17:19:25-07:00" level=info msg="192.168.220.1 - - [03/Jul/2026:17:19:25 -0700] \"GET /?rid=JZgWFhx HTTP/1.1\" 200 615 \"\" \"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36 Edg/150.0.0.0\""
2026-07-03 17:19:32	JZgWFhx	time="2026-07-03T17:19:32-07:00" level=info msg="192.168.220.1 - - [03/Jul/2026:17:19:32 -0700] \"POST /?rid=JZgWFhx HTTP/1.1\" 302 0 \"http://192.168.220.128/?rid=JZgWFhx\" \"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36 Edg/150.0.0.0\""

